

UniAssist, explained

The whole project in five pages · HCLTech Future Ready AI Engineer Hackathon · 6 October 2026

1. What UniAssist does

UniAssist answers students' questions about university rules and about their own records. Every answer:

- quotes the exact clause it comes from;
- carries a stamp saying what kind of answer it is;
- can be audited afterwards.

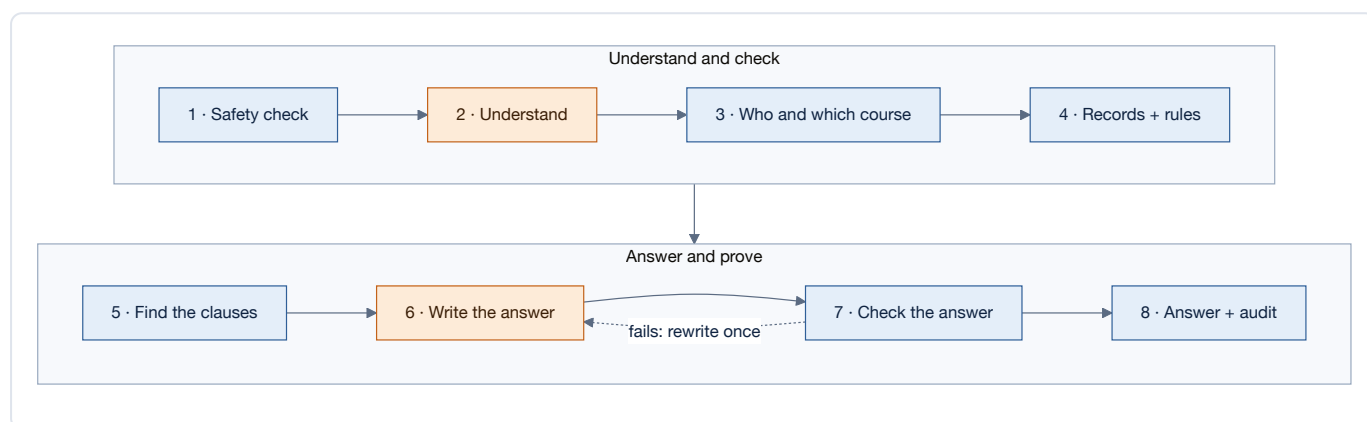
A student asks	UniAssist answers	Stamp
What is the minimum attendance for end-semester exams?	80%, quoting the circular that set it	As per rules
Am I eligible for the CS201 exam?	Not eligible: 77.50% is below 80%. Attend the next 5 classes.	Calculated
Is 65% enough if I have a medical certificate?	No: even with condonation the floor is 70%	As per rules
Which course? (<i>asked about "the exam"</i>)	Lists the student's courses to pick from	Which one?
Show me S1002's marks	Refuses: you can only see your own records	Refused
Is there a scholarship for studying in Antarctica?	"I could not find this information in the authorised sources."	Not found
Two circulars give different fees	Shows both, cites both, asks you to confirm with the office	Sources disagree

The one idea behind it: the AI model only understands the question and writes the explanation. Code decides everything that has to be right:

- who is asking;
- which rule applies;
- the arithmetic;
- the citations;
- the answer type.

A clever prompt or a planted sentence in a document therefore cannot talk it into a wrong verdict.

2. What happens when you ask a question



- 1. Safety check.** Blocks prompt injection, attempts to change the assistant's role, requests for other students' data and abuse. Phone numbers and emails are removed before anything is logged.

- 2. **Understand (AI, often skipped).** Works out what kind of question it is. When simple patterns already make it clear, code does this and the AI call is skipped.
- 3. **Who and which course.** The student comes from the sign-in, never from the question. If the question fits several courses, UniAssist asks which one.
- 4. **Records and rules.**
 - Exact maths over the student's records, such as 31 of 40 classes = 77.50%.
 - The threshold is read from the rule registry, never hard-coded.
- 5. **Find the clauses.**
 - Searches only documents in force on that date and for that programme, by meaning and by keyword together.
 - Replaced clauses are dropped.
- 6. **Write the answer (AI).** The model explains the verdict using only the clauses it was given. It has no tools and cannot change the verdict.
- 7. **Check the answer.** Every citation must be a clause it was given and every number must appear in a source. If the check fails, the model rewrites once; if it fails again, UniAssist quotes the clause directly.
- 8. **Answer and audit.** Code stamps the answer type and saves an audit record: sources, rules, tools, model, tokens and timings.

File c854b745

Am I eligible for the CS201 end-semester exam?
For S1002, Aaradhya Jain. Rules as of 6 October 2026.

CALCULATED
from your records

You are not eligible to sit the Data Structures (CS201) end-semester exam: your attendance is 77.50%, below the required 80%. Attending the next 5 classes would bring you to 36 of 45.

You need at least 80% attendance to be eligible for the end-semester exam, as stated in the circular from 2026-08-01. Unfortunately, your current attendance is 77.50%, which is below the required threshold.

Sources

1 Circular: Revised minimum attendance requirement
ACAD-2026-08, Section 1, page 1, version 1, in force from 1 August 2026

"1. With effect from 1 August 2026, the minimum attendance required to appear in the end-semester examination is raised to 80% in each course. This circular replaces clause 7.2 of the Academic Regulations (ACAD-REG-2024)."

Answered in 2.6 s, with 1 language model call. Every sentence of the answer is supported by the cited sources.

A personal answer. The verdict and the "next 5 classes" figure are computed by code; the model only wrote the explanation. The footer shows one AI call and that every sentence is supported by the cited source.

How this was decided	
Rules as of	6 October 2026
Rule applied	Minimum attendance: at least 80% ACAD-2026-08 Section 1, rule AUTO-ACAD-2026-08-ATT-1
Your record	CS201: 31 of 40 classes attended = 77.50% Not eligible for the CS201 end-semester exam: 77.50% is below the 80% minimum. Attending the next 5 classes brings you to 36 of 45 = 80.00%.
Sources that lost	Academic Regulations for B.Tech Programmes Section 7.2 says 75% Replaced by circular ACAD-2026-08 Student Help-desk FAQ 2026-27 Section 3 says 65% Lower authority: an FAQ ranks below a circular
Upcoming changes	None announced after 6 October 2026.

"How this was decided": the rule applied, the student's record, and why the 75% regulation and the 65% FAQ lost.

3. When documents disagree

Universities publish regulations, circulars, notices and FAQs, and they contradict each other. UniAssist settles this with the guide's precedence rules (Annex A), applied by code in five checks:

1. **In force and in scope?** The document must cover the date asked about and the student's programme and batch.
Future documents are shown as upcoming changes.
2. **Replaced?** A regulation or circular that says it replaces a clause wins over that clause.
3. **Higher authority wins:** regulation > circular > notice > FAQ. Unofficial posts never set a rule.
4. **Same authority:** the newer document wins.
5. **Still tied:** UniAssist says the sources disagree, cites both, and refers the student to the issuing office.

Source	Says	On 6 October 2026
Academic Regulations, section 7.2	75%	Replaced by the circular (check 2)
Circular ACAD-2026-08, section 1	80%	Applies
Help-desk FAQ	65% is enough	Loses on authority (check 3)
Student-council post	No minimum	Unofficial, never applies

Asked about 15 July 2026, the same question returns 75%, and the circular is listed as an upcoming change.

Context comes from the question too. UniAssist reads a date in the question ("As of 2026-12-10") and applies the rules in force then. "For B.Arch batch 2025" applies that programme's rules.

File a02e6cfa

What is the supplementary exam registration fee for B.Tech batch 2023 in October 2026?
General question. Rules as of 15 October 2026.

SOURCES DISAGREE
ask the issuing office

The sources disagree: Circular: Supplementary Examination Fee (section 1) says Rs 1200 and Circular: Supplementary Examination Fee (Finance) (section 1) says Rs 1500. Both have the same authority and date, so please confirm with Office of the Controller of Examinations, Aster University or Finance Office, Aster University.

Both sources have the same authority level and take effect on the same date, so the precedence rules cannot choose between them. Treat the value as unconfirmed until the issuing office says which one applies.

Sources

1 Circular: Supplementary Examination Fee
AST-FEE-2026-A, Section 1, page 1, in force from 1 September 2026

"1. The supplementary examination registration fee for B.Tech students of the 2023 batch is Rs 1,200 per course."

2 Circular: Supplementary Examination Fee (Finance)
AST-FEE-2026-B, Section 1, page 1, in force from 1 September 2026

"1. The supplementary examination registration fee for B.Tech students of the 2023 batch is Rs 1,500 per course."

Answered in 3.2 s, with 2 language model calls. Every sentence of the answer is supported by the cited sources.

Two circulars with the same authority and date give different fees. The answer cites both and makes no guess.

Adding a document happens while the system runs; the next question already uses it. The steps:

- check its metadata;
- skip exact duplicates;
- read the text, tables and scanned pages (OCR);
- split it by clause;
- flag any hidden instructions;
- turn stated thresholds into rule entries.

4. Safe and honest by design

Privacy and security

- **Your own records only.** Identity comes only from the sign-in, and the record tools have no "student ID" input. Another student's ID or name in a question is refused.
- **Input checks.** Prompt injection, jailbreaks, hidden or encoded instructions, bulk-data requests ("which students have a CGPA below 6.5") and requests to change records are blocked and logged.
- **Output checks.** No other student's ID, no system-prompt text, and no phone number or email that the sources don't contain.
- **Documents are data.** Instructions hidden inside an uploaded document are flagged and redacted, and never obeyed.
- **Rate limits.** Repeated attacks get the client temporarily blocked, and every block is recorded as a security event.

No made-up answers

- **Exact arithmetic.** 79.66% is never rounded up to 80%.
- **Citations.** Only the clauses retrieved for the question can be cited.
- **Numbers.** Every number must appear in a source, and comparisons are checked ("79.66% is below 75%" is caught).
- **Groundedness score.** Every answer is checked against its sources; weak drafts are rewritten or replaced with a quote.
- **Abstention.** If nothing relevant is found, the answer is "not found" rather than a guess.
- **No false "yes".** Lower-authority sources are hidden from the AI writer, so it can't repeat the FAQ's "65% is enough".

Fast and reliable

Technique	What it does
Code-first planning	Skips the planning AI call when patterns already tell the question type (about 1 s faster)
Caches	Repeat questions answer in milliseconds. Any new document or data load clears them, so an answer is never stale. Personal answers are never shared between students.
AI gateway	Retries, a circuit breaker and fallbacks. If the AI is down, rule, eligibility and refusal answers still work.
Token saving	Only the relevant sentences of each clause go to the AI, within a fixed budget, so it reads less and answers faster
Search	Meaning plus keywords, and student words mapped to official ones: "bunk" means attendance, "supply" means supplementary
Follow-ups	"What about CS202?" is understood from the previous question, for the same student only

5. The proof: data, tests and results

Synthetic data. No real student data is used anywhere.

- **Students:** 40 synthetic students across 2 programmes and 2 batches, written by the local AI model and validated by 16 automated checks.
- **Edge cases:** 14 set exactly. Examples: attendance exactly at 75%, one class short, a fail by one mark, and the 79.66% rounding trap.
- **Data card:** the full description is in `data/synthetic/DATA_CARD.md`.

Golden evaluation.

- **The set:** 146 questions, each with the expected answer type, facts, sources and tool results.
- **The buckets:** policy, procedures, versions and conflicts, personal records, other-student attempts, multi-step, unanswerable, prompt injection, follow-ups and caching.
- **The method:** each configuration ran on a fresh copy of the system.

Configuration (same code)	Correct	Citation accuracy	Hallucination	p50 / p95 time
A: simple baseline	77.8%	1.4%	2.1%	3.3 / 6.0 s
B: shipped	93.1%	89.3%	0.7%	3.3 / 6.1 s
C: B + reranker	93.8%	91.5%	0.7%	3.2 / 5.4 s

The evaluation traced each of B's failures to a cause, and every one was fixed. On the final code, B scores:

- **100% correct** (146 of 146, every bucket);
- citation accuracy 91.0% and a hallucination rate of 0.0%;
- about 1.4 AI calls per question, with a median answer time of 3.0 s.

C's reranker gained one question, so it stays an option, not the default.

Adversarial test pack. 13 synthetic documents for a fictional university:

- a baseline rule, a replacing circular and an exact duplicate;
- a false FAQ and unofficial posts;
- a future-dated rule and an out-of-scope programme;
- hidden prompt injection;
- tied fee circulars, a scanned notice and a keyword-stuffed trap.

All were uploaded through the real UI in a browser, and **13 of 13 checks passed live**. On top of that, 72 automated tests run offline.

6. Running it, and where things are

```
docker compose up --build -d          # UI http://localhost:8080 · API http://localhost:8000/docs
```

Local development: `uv sync`, then `uvicorn app.main:app`, then `python scripts/seed.py`. In `frontend/`, run `npm run dev`. The AI model is `llama3.1:8b`, running locally in Ollama, so nothing leaves the machine.

Folder	What is inside
<code>app/</code>	The API, the 8-step LangGraph pipeline, tools, search, ingestion, AI gateway, guardrails, caches
<code>frontend/</code>	The React app: Ask, Documents, Students and Audit pages
<code>data/corpus/</code>	The university documents, the source register and the curated rules
<code>data/synthetic/</code>	The student data generator, prompts, validator and data card
<code>eval/</code>	The golden set, evaluation runner, AI judge, report and adversarial test pack
<code>docs/</code>	Technical design, diagrams, audit samples and this guide

Who directed what (built with Claude Code, disclosed in `AI_USAGE.md`):

- **Manish Kumar:** the core build and its LangGraph orchestration.
- **Dhruv Kumar:** the production layer, data and evaluation.
- **Garv Bahl:** the frontend.